

Ritesh Kumar

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Career Objective

An aspiring Software Engineer with strong analytical and problem-solving skills, eager to contribute to impactful projects in real-world scenarios. Seeking a challenging role in the IT industry where I can apply my technical expertise and dedication to drive innovation and support organizational growth.

Education

Arka Jain University, Jamshedpur

Bachelor of Technology in Computer Science and Engineering (AI & ML)

CGPA: 8.35

Aug. 2022 – May 2026

Experience

Machine Vision and Intelligence Lab, NIT Jamshedpur

Research Intern

May 2025 – Present

Jamshedpur (On-Site)

- Working on a machine learning project for steel surface defect detection using real-world industrial datasets.
- Applying advanced computer vision techniques such as edge detection and feature extraction for accurate defect classification.
- Achieving up to 99.65% accuracy on a six-class steel surface dataset and 99.72% on the NEU-DET benchmark dataset.

Backend Intern

TechMantra Global

May – July 2024

Noida (Remote)

- Acquired comprehensive expertise in SpringBoot for backend development, skilled in designing and managing scalable applications.
- Developed responsive user interfaces with HTML, CSS, Bootstrap, and JavaScript to optimize user experience and engagement.
- Implemented RESTful APIs and integrated them with MySQL databases to ensure efficient data management.

Projects

Community AI Platform | FastAPI, React, SQLAlchemy, JWT

Live | [GitHub Link](#)

- Built an AI-powered platform to provide access to government schemes, learning resources, and job opportunities for underserved communities.
- Developed RESTful APIs using FastAPI and SQLAlchemy with JWT-based authentication and secure user management.
- Integrated AI models for multilingual chat assistance, personalized recommendations, and context-aware responses.
- Built modular async APIs with optimized database queries and Pydantic validation to deliver high-performance, secure backend services.

AI-Powered Customer Complaint Management System | FastAPI, React, PostgreSQL

Live | [GitHub Link](#)

- Built a full-stack web application for managing and resolving customer complaints using AI automation.
- Developed RESTful APIs with FastAPI and PostgreSQL, implementing JWT-based authentication.
- Integrated AI models for complaint classification, priority detection, and sentiment analysis.
- Created a responsive React dashboard for real-time complaint tracking and admin management.

Steel Surface Defect Detection | Python, CNN, Deep Learning

[GitHub Link](#)

- Engineered a Convolutional Neural Network (CNN) to classify steel surface defects into six categories: crazing, inclusion, patches, pitted surface, rolled-in scale, and scratches.
- Achieved 99.65% validation accuracy using an Attention-based Multi-Feature Fusion CNN (AMFF-CNN), leveraging attention mechanisms and multi-level feature integration to enhance model precision and generalization over conventional CNNs.
- Enhanced model generalization using data augmentation and optimized training with batch normalization and dropout.

Technical Skills

Programming Languages: C/C++, Python, JavaScript, HTML, CSS

Libraries & Tools: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, OpenCV

Frameworks: React.js, Node.js, Express.js, TensorFlow, Keras, Deep Learning

Developer Tools: VS Code, Git, GitHub, Jupyter Notebook

Databases: MySQL, MongoDB

Cloud & Platforms: Google Cloud Platform (GCP), GitHub Pages

Coursework: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Software Engineering

Interests: Machine Learning, Deep Learning, Web Development

Achievements

Shortlisted for **AutonomousHacks '26** Offline Finale among 2,000+ participants, qualifying for agentic AI hackathon organized by GDG Gandhinagar.

National Finalist, IBM Hackathon 2025 — Cleared IBM's HackerRank challenge and competed in a 24-hour national hackathon, demonstrating problem-solving, teamwork, and innovation.

CodeFest, Arka Jain University — Secured **2nd place** during the Engineering Day competition.

Earned a **Certificate in Database Management Systems** through NPTEL.

Organized "Hack Horizon 2K25" — A university-level hackathon fostering innovation, collaboration among tech enthusiasts.

Strengths

Analytical Thinking — Skilled in identifying patterns, interpreting data, and solving problems logically.

Critical Reasoning — Strong ability to assess complex issues and make effective decisions.

Teamwork & Collaboration — Excellent at working within diverse teams to achieve shared goals.

Time Management — Efficient in prioritizing tasks and meeting deadlines consistently.

Professional Attitude — Maintain an optimistic and adaptable mindset fostering continuous learning.

Leadership & Extracurricular Activities

Serving as Class Representative (CR) — Currently acting as a bridge between faculty and students, ensuring effective communication and academic coordination.

Community Lead, GDG on Campus AJU — Actively contributing to the growth of the developer community by organizing tech events, workshops, and peer-learning sessions.

Team Member, Gate Club, Arka Jain University — Engaged in organizing workshops and discussions to support GATE exam preparation and foster peer learning among members.

Club President, Code & Compute Society (CCS), Arka Jain University — Leading club operations, coordinating with faculty and university officials, and driving initiatives that promote coding culture and technical excellence on campus.